

Designing and Installing a Fiber-fed spectrograph in the Pupin Rutherford Dome for Multi-Object Spectroscopy

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ABSTRACT

We discuss the design and installation of a fiber-fed spectrograph for the Celestron EdgeHD 14 telescope in the Pupin Rutherford Dome. Building upon the newly installed field de-rotator, a multi-fiber design would allow for the simultaneous spectra acquisition of multiple celestial objects. A custom aluminum plug plate, drilled to fit and position optical fibers at the focal plane is used to feed light from up to 9 objects into a bench spectrograph. For observational ease, an additional imaging fiber is factored into the plate design to provide real-time field resolution. When mapping the celestial field onto the plate, the cartesian positions were found through a gnomonic projection. To demonstrate usage of a multi fiber design, we modeled the system off of Melotte 111, an open cluster that targets cluster members, galaxies, and field stars. We discuss final results through calibration efforts using white and neon light, and on-sky testing.

1. TECHNICAL BACKGROUND

The current instrument that is used in the Pupin Rutherford dome for collecting single source data is a self-guided spectrograph built by the Santa Barbra Instruments Group. Spectroscopy offers deep insights into a wide range of astronomical research questions such as stellar properties, cosmic velocity of objects, exoplanet characterization, and much more. However, the instrument we currently use can only obtain spectra of singular sources. Multi fiber systems address this issue, offering the capability to simultaneously measure the spectra of multiple sources.

Current systems that utilize this technology work with bundles of hundreds to a thousand fibers to perform large-scale all sky spectroscopic surveys. These plug plate systems have been further evolved into robotic Focal Plane Systems (FPS). Key examples of robotic positioner usage are with the Sloan Digital Sky survey (SDSS) and the Dark Energy Spectroscopic Instrument (DESI). The idea here is to reduce manual error for varying fields that require frequency X, Y repositioning (Owen et al. 1994). Of course with our small set up, our design goals would be met with a simple aluminum plate with manually adjusted fibers for a single field.

A key consideration in a fiber-fed spectrograph is the fibers themselves, acting as both the focal plane of the telescope and the entrance slit of the spectrograph. As such, their core diameters have direct effects on spectral performance and geometric matching to the star field. Our system constraints dictate a 1-inch diameter operating spectrograph input paired with 100 micron fibers, necessitating precise plug-plate packing, tolerances, and multi-object positioning within the input field. For the Celestron EdgeHD 14, with a focal length of $f = 3910$ mm, we get a plate scale of $p = 206265/f \approx 52.75$ arcsec/mm. Given our target field is a 15 arcminute portion of Melotte 111, we get a $\frac{15 \times 60}{p} \approx 17$ mm footprint onto the focal plane. This leaves us with plenty of room to station our 100 micron fibers. One final consideration is the angular field of view of the fibers themselves. For our 100 micron fibers on a plate scale of 52.75 arcsec/mm, their fields of view are $\theta = \frac{100}{1000} \times 52.75 = 5.28$ arcseconds. This enables us to adequately isolate point sources on our star field, while maximizing the signal-to-noise ratio due to urban glow or other light pollution.

2. METHODS

2.1. Plug Plate Calculations

Before any initial images of our field were captured from the Pupin roof, we were able to access the rough field of view using the Aladin Sky Atlas. Aladin conveniently superimposes catalogs and databases such as the SIMBAD database, allowing us to utilize the right ascension and declination of celestial objects in the projection (Fig. 1).

As mentioned, we performed a gnomonic (tangent-plane) projection to calculate the specific (X, Y) position for each fiber using the sky coordinates of the target objects. In this projection, a flat plane is placed tangent to the celestial sphere at the field center and each target is projected onto that plane geometrically (Weisstein 2024). In our case, that tangent plane represents plug plate fiber position. For a target at coordinate (α, β) with field center (α_0, β_0) , the distance is defined as $\Delta\alpha = \alpha - \alpha_0$ and the projection denominator is,

$$D = \sin \delta_0 \sin \delta + \cos \delta_0 \cos \delta \cos(\Delta\alpha), \quad (1)$$

The tangent-plane offsets ξ (East-West) and η (North-South), both in radians, are then

$$\xi = \frac{-\cos \delta \sin(\Delta\alpha)}{D}, \quad \eta = \frac{\cos \delta_0 \sin \delta - \sin \delta_0 \cos \delta \cos(\Delta\alpha)}{D}. \quad (2)$$

All that's left is to compute are the physical millimeter spacial coordinates that are attained using the plate scale. For the Celestron EdgeHD 14, we know the focal length to be $f = 3910$ mm and thus we can use this to get the plate scale, $p = 206265/f \approx 52.75$ arcsec mm⁻¹. The final plate positions are

$$X = \frac{206265 \xi}{p}, \quad Y = \frac{206265 \eta}{p}, \quad (3)$$

where the value of 206265 appears as a conversion from radians to arc seconds.

Alongside the plate calculations, preliminary observations were made to view our field from the roof. Multiple frames were taken with 2 minute exposures on the ZWO ASI2600MC Duo. We noted that galaxies appear as extremely dim objects where many cluster members were bright and visible.

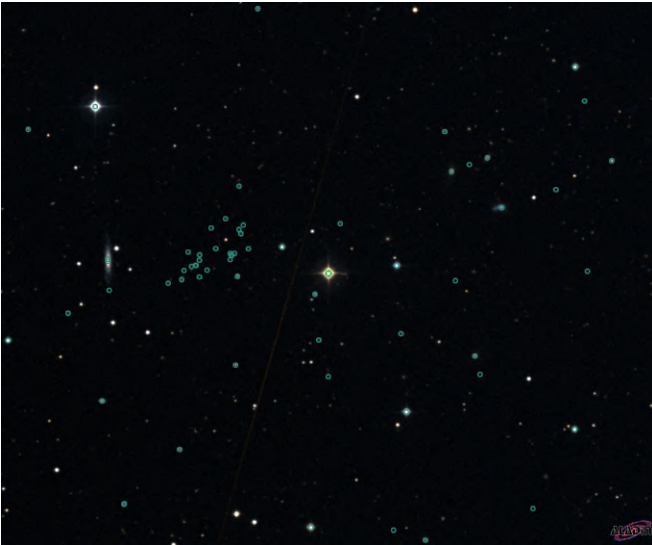


Figure 1. Cut out of objects cataloged by SIMBAD in approximate FOV

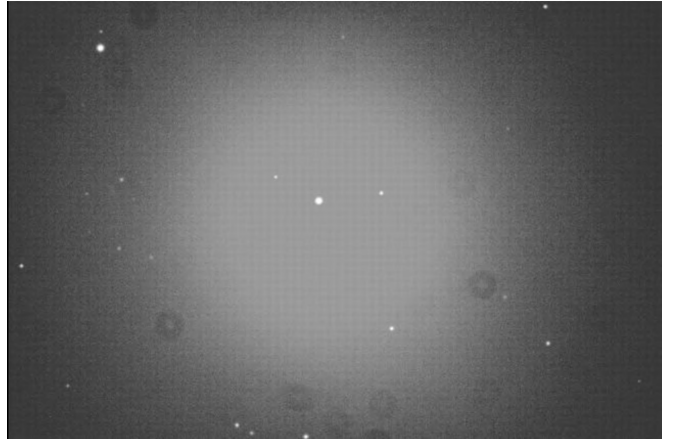


Figure 2. Open star cluster Melotte as seen from the Pupin dome

Table 1 lists the (X, Y) positions for each target on the plug plate, as output by the gnomonic projection script. The coordinates are given in millimeters relative to the field center, which is additionally aligned with the brightest star BD+26 2319, and the imaging fiber is placed with a bright member of Mel 111. The column r gives the radial distance from the plate center, and Sep denotes the angular separation from the field center on the sky in arc seconds.

Table 1. Computed plug plate hole positions for the Melotte 111 field, derived from gnomonic projection of SIMBAD coordinates onto the focal plane of the Celestron EdgeHD 14 ($f = 3910$ mm, $S = 52.75$ arcsec mm $^{-1}$). The field center is BD+26 2319 at RA = 183.1257°, Dec = 25.2755°.

#	Object	X (mm)	Y (mm)	r (mm)	Sep (")	Type
0	Center: BD+26 2319	−0.000	0.000	0.000	0.0	guide
1	Z 128-53	−7.725	0.418	7.737	408.1	galaxy
2	Mel111 AV (1)	−1.625	0.920	1.868	98.5	star
3	Mel111 AV (2)	2.386	0.273	2.401	126.7	imaging
4	Mel111 AV (3)	−8.172	5.825	10.036	529.4	star
5	Mel111 AV (4)	2.732	−4.844	5.562	293.4	star
6	Mel111 AV (5)	8.625	−5.448	10.201	538.1	star
7	2MASS J12125477+2514417	−6.329	−2.083	6.663	351.5	star
8	SDSS J121242.56+25	−3.187	−0.109	3.189	168.2	star

2.2. Distortion Calculations

Because of the precise spacial mappings, we additionally performed distortion calculations on the current instrument assembly on the Celestron EdgeHD. The convention we followed was developed by the Spitzer Science Center (SSC) for distortion generated by scan mirror position. The distortion is represented as polynomials where the coefficients are found in the FITS headers of each Basic Calibrated Data (BCD) product (Shupe et al. 2005).

To perform the extraction we used simple script that selected the corresponding header page. The non-zero coefficients are listed in Table 2. It was ultimately concluded these values were subtle enough that we did need not factor them into the plate calculations.

Table 2. Non-zero SIP distortion coefficients extracted from the DSS FITS header for a previous observation of M82 on the telescope.

Polynomial	Coefficient	Value
<i>Forward, axis 1 (A)</i>		
	$A_{0,2}$	4.168×10^{-7}
	$A_{1,1}$	5.445×10^{-7}
	$A_{2,0}$	3.958×10^{-7}
<i>Forward, axis 2 (B)</i>		
	$B_{0,2}$	-1.404×10^{-6}
	$B_{1,1}$	8.534×10^{-7}
	$B_{2,0}$	-8.793×10^{-7}
<i>Inverse, axis 1 (AP)</i>		
	$AP_{0,0}$	-2.736×10^{-5}
	$AP_{0,1}$	-6.446×10^{-7}
	$AP_{0,2}$	-4.169×10^{-7}
	$AP_{1,0}$	1.938×10^{-7}
	$AP_{1,1}$	-5.452×10^{-7}
	$AP_{2,0}$	-3.954×10^{-7}
<i>Inverse, axis 2 (BP)</i>		
	$BP_{0,0}$	-1.742×10^{-3}
	$BP_{0,1}$	4.179×10^{-6}
	$BP_{0,2}$	1.405×10^{-6}
	$BP_{1,0}$	-1.452×10^{-6}
	$BP_{1,1}$	-8.565×10^{-7}
	$BP_{2,0}$	8.825×10^{-7}

2.3. Plug Plate Assembly

Using the spacial mappings from Table 1, CAD models of both the Plug plate and the spectrograph input were designed and exported as DXF files for manufacturing. The plug plate was cut from 1.5 mm thick aluminum into a 2 in. disc. Similarly, the spectrograph input plate was cut into a 1 in. aluminum disc. For the input, we staggered the fibers to act as a traditional slit at the spectrograph entrance.

Initial fabrication was attempted using a water jet cutter in the Columbia Makers Space, however the positional accuracy of the water jet was found to be insufficient. The deviations were large enough that fiber holes with small separation would overlap. As a result, we transitioned to a drill press which allowed for precise 1.25 mm punches.

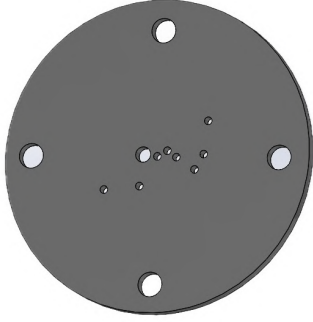


Figure 3. Focal plane plate, 2" diameter

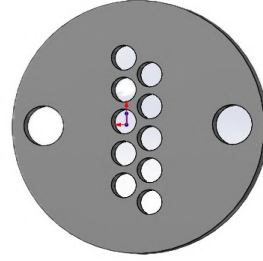


Figure 4. Staggered spectrograph input plate, 1" diameter

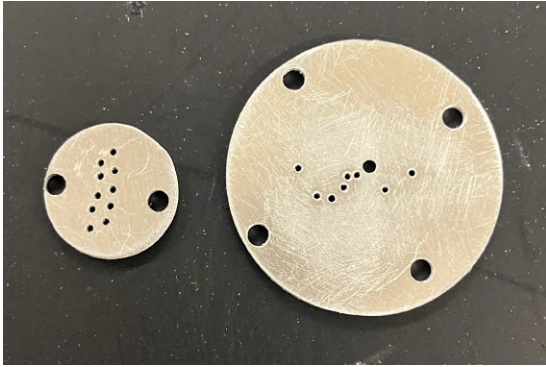


Figure 5. Aluminum plates drilled using a manual drill press

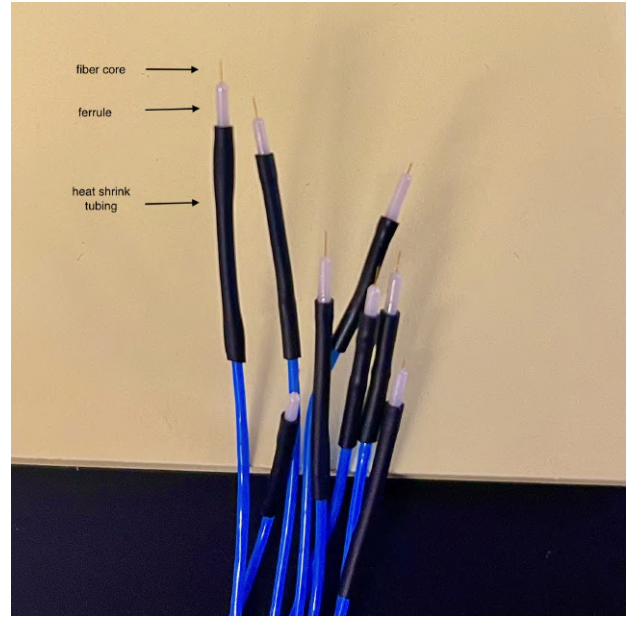


Figure 6. Fibers with ferrule attachments

2.4. Optical fiber assembly

The optical fibers we used for the assembly of this instrument were 100 μm in core diameter, threaded through ferrules with a total outer diameter of 1.25 mm. Each fiber was cut to 2 m in length to ensure there be enough reach from the telescope focal plane to the bench spectrograph. After measuring out 9 equal length fibers, they were individually threaded through rubber insulated tubing.

A ferrule was then applied to each end of the fibers and secured with a small bead of epoxy. Once the epoxy had set, excess length was carefully cleaved to ensure fibers were of equal length outside of the ferrule for focal alignment (see Fig. 6).

To polish the fibers, we cycled through three films of varying grit strength beginning with $12\ \mu\text{m}$ to $3\ \mu\text{m}$ and finally $1\ \mu\text{m}$, each for about 1 minute. Finally, to ensure no slipping from the outer layer of insulation and complete fiber coverage, we applied heat shrink tubing on both sides of the fibers, partially covering the ferrules. When actually installing the fibers in both plates, measurements were taken from the edge of the ferrule to the optical fiber. The target value for the remaining fiber sticking out of the ferrule was meant to be $6/16''$ and this was achieved through ruler and visual estimates while cleaving away excess.

2.5. Optical Design and fiber mounting

Although the tolerance of the fibers to the plug plates was sufficient to allow the fibers to remain plugged without slippage, we applied a silicone rubber RTV sealant to the back-facing sides of each plate. To allow for flexibility, especially near both plates, the fibers were not bundled along their entire length (as seen in Fig. 7) and rather were left to freely travel with directional changes made to the telescope mount. The drawing below is the envisioned design for the instrument, where the $1.25\ \text{m}$ fibers transmit light from the objects directly to the tabletop spectrograph along a $1.25\ \text{m}$ path. As for guiding and field positioning, the external imaging fiber is directly oriented towards a CMOS detector. The simple tabletop spectrograph design consists of a collimator lens, a grating, a focusing lens, and a detector. The installed spectrograph is shown below in Figure 8. Not pictured is an additional mounted detector where the imaging fiber is fed.

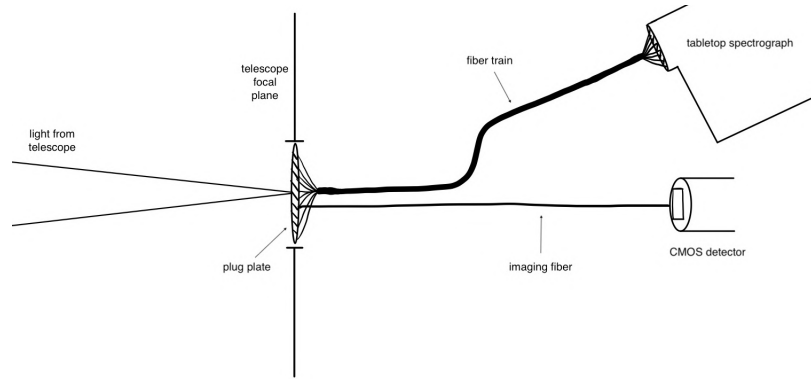


Figure 7. Fiber train schematic

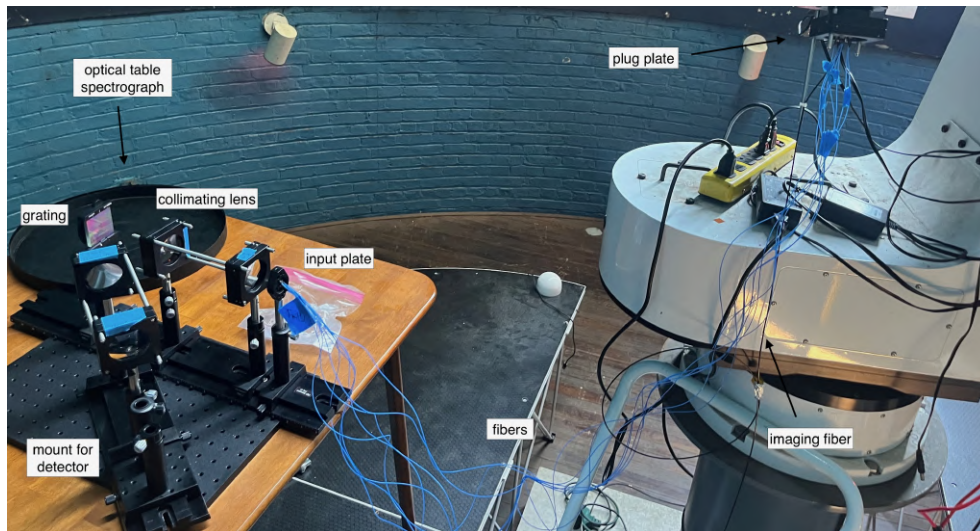


Figure 8. Installation of fibers and plug plates

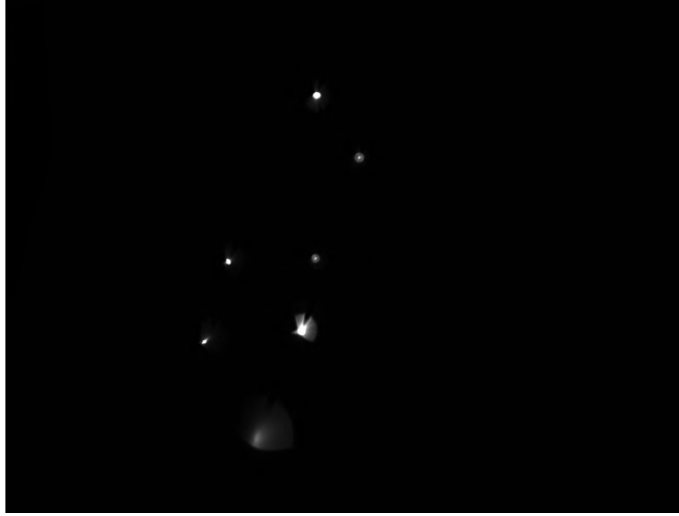


Figure 9. Spectrograph plate input test, zeroth order. Three fibers are generally in focus with on another, while the rest are blurred because of focal plane misalignment. 7 functional fibers are currently plugged into the plate

3. RESULTS AND ANALYSIS

After initial attempts of on-sky observations using the bright star Arcturus, we conclude that due to a lack of design precision, this current spectrograph is not able to attain multiple spectra from a deep sky object. The telescope was equipped with a flip mirror to alternate between directing light to an imaging camera and to the spectrograph fiber feed. While tracking Arcturus, only a single point source was visible, suggesting that light signal was successfully reaching at most one fiber during the observing session. To further investigate whether adjacent fibers were not transmitting light, a spiral offset pattern was cycled through in increments of several arc minutes to scan across the plate in an attempt to reach other fibers. After several attempts, no additional fiber illumination could be detected during this observing session.

We assume design factors likely contributed to this result with the most probable cause being a misalignment between ferrule/fiber position at the focal plane. Before observing on the sky, light from a lamp was applied to the fiber openings in the plug plate focused through moving the detector mount back and forth. Three fibers were chosen as the primary focus and the rest of the fibers were slightly unfocused, indicating a focal plane offset of around 3 – 5 mm (see Fig. 9). Related to the issue of fiber alignment, throughput loss from the fiber polishing process and focal ratio degradation may have further reduced the signal to undetectable levels.

3.1. Fiber Testing and Calibration

While on-sky observing was not possible, we can speak towards the calibration measures taken using a white light and neon lamp to map pixel to wavelength through the dispersion solution. As mentioned before, key features of this spectrograph include a grating with ≈ 600 lines/mm, 50 mm lenses, and a camera focal length of ≈ 300 mm. The grating equation can be used to specifically calculate the angle of diffraction using,

$$d(\sin \theta_i + \sin \theta_d) = m\lambda, \quad (4)$$

where m is the diffraction order, λ is wavelength, and $d = 1/600$ mm is the grating groove spacing that comes from the equation $d = 1/\# \text{ of lines}$. A key feature here is that grating tilt angle corresponds to diffraction order. For example, $m = 0$ essentially acts as a mirror with no diffraction. Further steps can be taken to relate physical position on the detector to wavelength, however, we can also determine the dispersion of the wavelength through comparison to well-known emission lines. In this case, we use a provided set of lines at precisely known wavelengths (MDM Observatory 2024). The dispersion relation is then a polynomial fit (where in this case we will just use a linear fit) that can be matched to the raw spectrum. This first order linear fit is given by

$$\lambda(x) = \lambda_0 + c_1 x, \quad (5)$$

where x is the pixel position along the dispersion axis, λ_0 is the zero point wavelength, and the coefficients are determined by a least squares fit. A python script was used to roughly orient the known emission lines to our data.

The staggered slit orientation that we used for the input plate causes the spectra to be offset by certain X, Y values. Due to these offsets, a 1D emission extraction of the entire file gave a smeared indistinguishable result. Instead, isolated bands of spectra were selected by isolating pixel ranges along the Y axis. For example, taking our data from the neon lamp with no filters applied (see Fig. 10), we were able to isolate the central most illuminated lines to perform a best fit. In Fig. 11, two fibers are estimated to lie between $y = 830 - 970$ and $y = 700 - 800$. Additionally, the headers for Fiber A and Fiber B display the dispersion relation with the wavelength centers (λ_0) and the reciprocal linear dispersions (c). The resulting reciprocal linear dispersions were $c_A = 8.517 \text{ nm/mm}$ and $c_B = 8.475 \text{ nm/mm}$, with respective central wavelengths $\lambda_0 = 5537.7$ and 5558.0 angstrom. Converting this dispersion relation comes from considering the ZWO ASI1600 detector which has an effective pixel size of $7.6 \mu\text{m}$ with bin size of 2.

Due to the plate offsets, it makes sense that solutions share close to the same c value with differing central wavelengths. It is also important to note from the first order spectra, the aberrations here in emission spikes are due to slight spectrograph misalignment, while throughput corresponds to losses in strength from the fiber fabrication process.

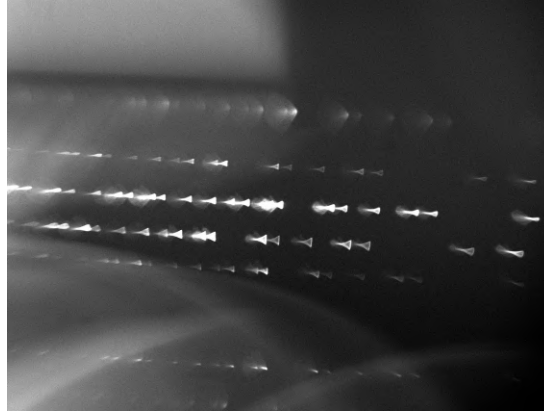


Figure 10. Emission spectrum of neon lamp across 7 fibers to the 1st order

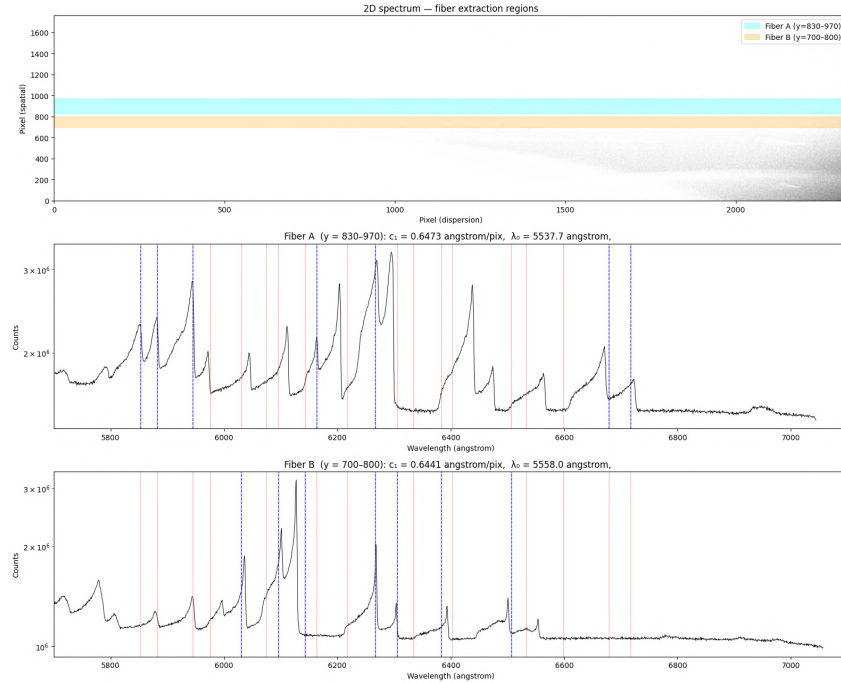


Figure 11. (Top) Extraction regions of two bright fiber output. (Middle and Bottom) Emission spectra of two fibers overlaid with known neon spectrum where blue lines correspond to estimated matched neon peaks and red lines are all known neon lines

Other images that provided valuable insights about fiber efficiency are shown below. Combinations of spectral images were taken using white light and the neon lamp with and without an H-alpha filter. While these were not considered in dispersion calculation, we can confirm that 5-7 fibers are transmitting the same light simultaneously. Further comments are below.

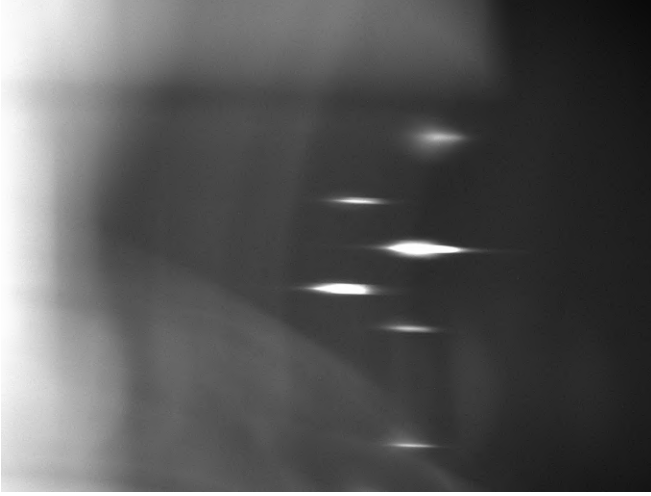


Figure 12. Initial H α filter on white light, serving as a position reference for the $\lambda \approx 656$ nm, $\Delta\lambda \approx 8$ nm filter. Images were repeated (Fig. 13) due to saturation

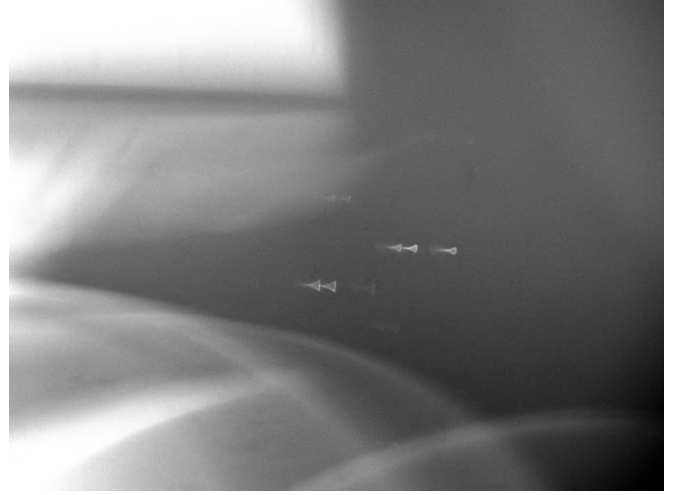


Figure 13. Neon lamp spectrum through the H α filter, isolating neon emission lines within a ~ 652 – 660 nm band. It is good here that we see similar line profiles across multiple fibers that are consistent with the known neon lines at 659.9 nm identified in the full dispersion solution.

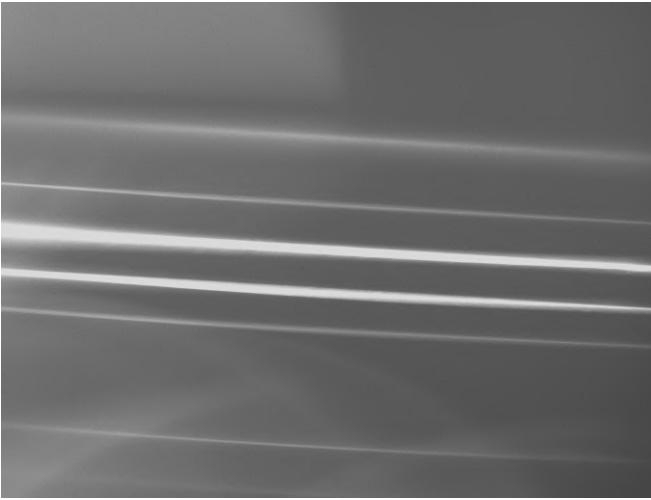


Figure 14. White light exposure. Continuous fiber traces are visible spanning the full detector width, with brightness variations between fibers likely reflecting differences in throughput due to fiber polishing and alignment.

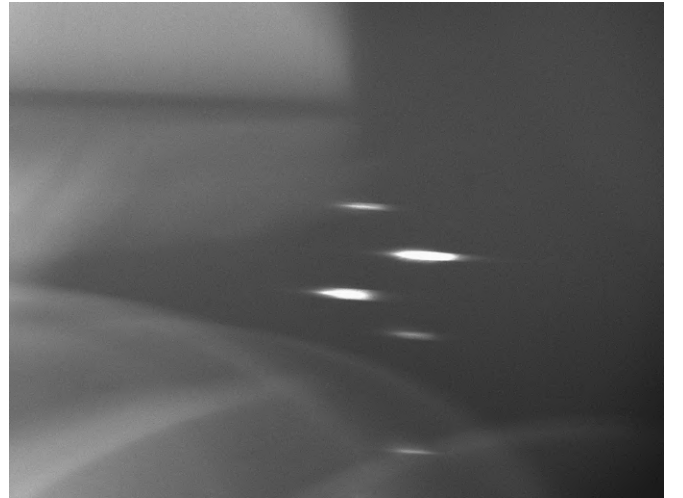


Figure 15. White light through the H α filter, isolating a narrow wavelength segment of the continuum.

4. CONCLUSION

Our efforts to develop and install a fiber-fed spectrograph for the Pupin Rutherford Dome demonstrate the feasibility of adapting multi-object spectroscopic techniques for future observation. Through the design of both the focal plane custom plate and the spectrograph input plate, implementation of gnomonic projection calculations, fabrication of multiple fibers, and calibration on a table spectrograph, this project establishes a framework for multi-fiber spectral

measurements. Testing in a lab setting with white light and neon lamps confirms that we were successfully transmitting light from 5 – 7 fibers. Further calibration results using the neon lamp confirmed that the multiple spectra shared a linear dispersion relation of approximately 8.5 nm/mm.

Unfortunately, the current instrument configuration is not yet capable of successfully obtaining multiple spectra through on-sky measurements, however, the calibration and testing processes reveal the several limitations that could inform future improvements. We primarily identify that issues stem from focal plane and ferrule alignment precision. When focusing the fibers on the spectrograph, we see that there are millimeter-scale offsets that prevent bringing all fibers simultaneously into complete focus. These drawbacks of course translated to on-sky attempts where only one fiber from the plug plate was successfully transmitting a detectable signal. Despite these challenges, the successful calibration and multi-fiber transmission demonstrate that the design concepts used in the project are viable for further precision implementation efforts.

For recommended future iterations, we would strongly consider incorporating drilling techniques that require a higher machining tolerance (potentially a high-speed CNC mill) and refined focal alignment procedures, specifically focusing on ferrule alignment.

Finally, we would also like to sincerely thank Professor Schiminovich for his time, resources, and continual guidance throughout every stage of this project.

5. CONTRIBUTION

This project required a joint effort from all group members in fabricating the fibers, on sky observing, and fiber testing/calibration. Calculations and report writing were done primarily by Stella with edits from Nathaniel.

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